Finite 2-Variation Jordan Curves and Inscribed Rectangles via Asano–Ike

Internal proof package — clean-room verified (Directive 009)
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Abstract

Jordan curves of finite 2-variation inscribe every prescribed rectangle, via simple polygonal interpolation [SOURCE], the simple-arc area inequality T021 [LOCAL], primitive compactness, frozen T018 smoothing [LOCAL], and Asano–Ike Theorem 1.1 [SOURCE]. Internally verified only.

1 Scope

Finite 2-variation only; not all Jordan curves. Critical $p = 2$ modulus covered separately (frozen T020).

2 Simple Jordan interpolation [SOURCE]

Boedihardjo–Geng Theorem 2.2 yields simple chord polygons $P_n = c^{D_n}$ with mesh $\to 0$.

3 T021: Simple-arc area bound [LOCAL]

Theorem 3.1 (T021)**.** $|\mathcal{A}_\alpha(s,t)| \le \frac{\pi}{4}|\alpha|^2_{2\text{-}var;[s,t]}$ *for simple polygonal arcs.*

Chord–lobe decomposition; isodiametric bound; degenerate limits by subdivision.

4 Local 2-variation control [LOCAL]

$\omega_c(s,t) = |c|^2_{2\text{-var};[s,t]}$ is superadditive and continuous. PL variation controlled at partition nodes.

5 Primitive equicontinuity and Arzelà–Ascoli [LOCAL]

$f_n(t) - f_n(s) = \frac{\xi_n(s)+\xi_n(t)}{2}(x_n(t) - x_n(s)) - \mathcal{A}_{P_n}(s,t)$. Extract $f_{n_k} \to f$ uniformly.

6 T018 smoothing [LOCAL]

Velocity-matched cubic corners; adaptive tolerance for diverging polygon length (frozen release).

7 Lift and Asano–Ike [SOURCE]

Period-count estimate; Theorem 1.1 and Remark 5.6.

8 Limitations

Depends on Asano–Ike preprint. Infinite 2-variation not covered (T023 open).